A Minimal Epistemic Core

Differentiation, Receptive Dependence, and Registration

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ABSTRACT. For a thing to appear in any respect, it must bear a difference, and another difference must hold in virtue of that one. This paper proposes these two conditions, differentiation and receptive dependence, as the minimal core of perception and of whatever knowledge rests on it, since nothing is perceived that does not appear. It states them between actual terms only, without alternatives, order, representation, or a knowing subject, and defines registration, the exact form of appearing, as one difference holding in virtue of another. What a subject adds lies outside the core.

1. Introduction

Theories of knowledge ordinarily begin with a subject who believes, judges, or is justified, and ask under what conditions such states amount to knowledge. This paper asks what must hold before that: what is required for anything to appear at all.

Its answer has two parts. There must be a difference in what is under consideration, since nothing appears where everything is alike. And there must be another difference that holds in virtue of the first. The paper calls this structure *registration*: to appear is to be registered.

To appear is to show up. The word is not used in the sense of seeming, as opposed to being, and it implies no subject to whom something appears. Perception and knowledge, which involve a subject, are not treated here; nothing is perceived that does not appear, so the conditions are necessary for them. The conditions are formal: they are stated without reference to space, time, matter, or causation, and examples from particular domains show how they are met there without fixing what they are. They are necessary conditions, not a sufficient account, and the assessment of judgement lies outside their scope.

The exposition uses ordinary logical resources—identity, distinction, relation, and inference—to state its commitments. This does not attribute those resources to what the commitments describe, and in particular not to a knowing subject.

A distinction drawn in the exposition is not thereby a distinction drawn by its object.

The proposal has close neighbours. Dretske (1981) analyses the information carried by a signal in terms of nomic dependence and conditional probability, and separates it from meaning and belief. Floridi's diaphoric definition of data takes a lack of uniformity—a difference—as the most elementary constituent of information (Floridi 2011), and his informational structural realism holds that the relation of difference is constitutive of its relata (Floridi 2008). Smith's metaphysics of presence makes *registration* its central notion, as the way in which a world comes to be taken as one of objects (Smith 1996).[1] Enactivist accounts describe cognition as the structural coupling of a system with a medium, both of which belong to a larger organization (Maturana and Varela 1987; Thompson 2007). The present account differs from each in three respects. It separates the difference that is registered, the difference that registers it, and the dependence that connects them. It states that dependence between differences of actual terms, without probabilities, causes, or alternatives on either side. And it distinguishes the direction of dependence from any order of occurrences.

Sections 2 and 3 concern the difference in what is under consideration. Sections 4 and 5 concern the dependence and the receiving side. Section 6 defines registration, Section 7 gives two cases, and Section 8 concerns order.

2. Differentiation

Let F designate what is under consideration. Calling it a *field* fixes the scope of the inquiry; it implies neither a spatial region, nor a collection of discrete objects, nor something already presented to a subject. Nor does the notation establish a boundary intrinsic to reality: F identifies what the account concerns, and any claim about its actual boundaries would require separate justification.

(D) differentiation

For anything of F to appear, F must bear a difference.

Nothing appears where everything is alike: to appear is to stand out from something. If F is taken whole, it appears by its difference from something beside it, and what is under consideration is then the two together. If F is everything, there is nothing beside it, and it appears only through the differences within it. The same condition holds of anything said of F: to say something of F is to identify it, and to identify is to set apart.

1. The present use of the term is narrower. Smith's registration is performed by subjects and yields objects; registration here presupposes neither and concerns only the dependence of one contrast on another.

The notion of a difference is not defined, because any definition would use it: to define something is already to set it apart from something else. It can be pointed to. Take anything that is not alike throughout. Something in it is one way, and something in it is another way: there are two, and in some respect they are not alike. The account calls this a *contrast*. The two that differ are its *terms*, that in which they differ is its *respect*, and what has the two terms *bears* the contrast. A contrast is always in some respect and always between two terms; a difference among more terms is several contrasts. Two contrasts are the same when their terms and their respect are the same, so the same two terms may bear two contrasts, in two respects.

Both terms of a contrast are actual. The account uses no difference between something actual and an alternative, that is, a way that nothing actual is. An alternative would have to be held somewhere: in a stock of possibilities, in a repertoire belonging to whatever receives, or in the description that specifies it. The account admits none of these into the core. The core therefore does not provide for three things to appear: which way something is as against a way that nothing is; anything of a term that differs from nothing actual; and that two terms are alike. If they appear, they require more than the core contains. "Actual" is meant without tense.

That F bears a contrast says that there is a difference, not that F could be differentiated. It says nothing of which contrasts F bears, of whether one differentiation of F is privileged, or of whether what differs is objects, qualities, regions, or values. It implies nothing about change or order. And it involves no knowing subject: that F bears a contrast does not imply that anything differentiates it.

The contrasts concerned are those of F itself. The account concerns a differentiation attributable to its target, rather than one supplied entirely by an unrelated description. A difference that holds, even in part, because the account has specified it belongs to the description, and it is left out.

3. Articulation

An account speaks of a particular contrast of F by means of an *articulation*, written

$$\langle a, \delta, b \rangle_F,$$

which names two terms, a and b, and a respect, δ. The articulation is *specified* by the account, not by F. It is *realized* when a and b are actual terms of F and differ in δ. Specification selects a contrast; it does not create one, and it can fail: a name may pick out nothing, or the terms may not differ in the respect named. The order in which the terms are written, their labels, and their grouping belong to the articulation, not to F. Specification fixes what an inquiry concerns, as F fixes its scope; it does not supply what the inquiry finds.

The terms of a realized articulation are both actual. They are not alternatives of which one is the case and the other not; the contrast is the difference between two terms that are both actual. The schema does not assume that a and b exist before the contrast as independent objects.[2]

Two misreadings of the schema should be excluded. First, δ is not a third item situated between a and b. To treat it as a separator would presuppose spatial organization, whereas δ is the respect in which the two differ. Second, b need not comprise everything that remains once a has been selected. That reading would require an exhaustive partition of F, and a contrast between two terms is not equivalent to a division into "this" and "everything else within the field": the latter adds the requirement that every case within the scope fall on one side or the other (Figure 1).

A realized articulation remains an articulation of F. Nothing so far implies that its contrast appears anywhere.

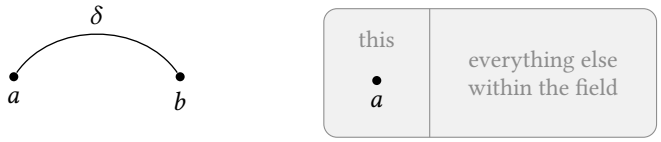
4. Receptive Dependence

A contrast of F and a contrast borne elsewhere are two facts, and nothing said so far connects them. They could be unrelated. A further commitment is therefore needed.

> (C) receptive dependence
>
> *Something of F appears only where another contrast holds, at least partly, in virtue of a contrast of F.*

"In virtue of" is not defined. It is the notion in use whenever one thing is given as the reason why another is so. Two things may both be the case without either having anything to do with the other; or one may be the case because the other is. (C) says that appearing is of the second kind. It is receptive: where something of F appears, something is as it is because of how F is. The account also calls this *dependence*. In what follows K is a contrast of F and K' a contrast that holds in virtue of it.

2. That a difference need not be grounded in terms identified before it is argued by Deleuze in his critique of the subordination of difference to identity (Deleuze 1994, ch. 1); the present account leaves the question open. Deleuze also rejects the possible as a mere image of the real, and the present account likewise uses no stock of possibilities.



(a) contrast (b) exhaustive partition

FIGURE 1. Two constructions over F. (a) An articulation $\langle a, \delta, b\rangle_F$: a and b mark terms, both actual, and δ, drawn as a relation rather than as an item between them, is the respect in which they differ. (b) An exhaustive partition, which adds the requirement that every case fall on one side or the other. Only (b) is drawn as a region, because only (b) carries the extensional commitment that a region suggests; in (a) the placement of the marks has no spatial meaning.

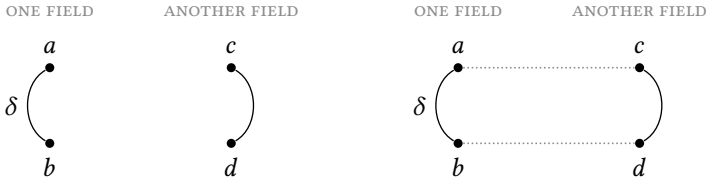
The notion is used under three conditions. It relates contrasts that both hold. Nothing holds in virtue of itself. And it has a direction: that K' holds in virtue of K does not bring with it that K holds in virtue of K'.[3]

Dependence is neither resemblance nor pairing. Two contrasts may be alike, and the terms of one may be paired with the terms of the other, without either holding in virtue of the other (Figure 2).[4] Both relations can be read in either direction, and neither ties a contrast to one other contrast rather than to any that is like it. The same holds where two contrasts agree because both depend on a third: each depends on the third, and neither on the other. (C) requires none of these, and in particular no pairing of terms.

Nor is dependence the same as what would have held otherwise. K' may hold in virtue of K although, had K not held, something else would have brought K' about; and two contrasts may be such that neither holds without the other although both depend on a third. What would have held otherwise is evidence of dependence, not what it consists in.[5]

The dependence may be partial: K' may hold in virtue of K and of other things as well, among them how the field that bears K' is.[6] Bateson's definition of information as "a difference which makes a difference" compresses the whole structure into one phrase (Bateson 1972); MacKay had earlier described informa-

3. The first and third features are familiar from the literature on metaphysical ground (Fine 2012). The account does not identify the dependence with ground.
4. Goodman (1976, ch. 1) makes the parallel point about representation: resemblance is reflexive and symmetric, whereas representation is neither, so resemblance cannot be what representation consists in.
5. Counterfactual analyses of dependence require alternatives to the actual case. Where contrasts hold necessarily, they meet a further difficulty: on the standard semantics, counterfactuals with impossible antecedents are all vacuously true (Lewis 1973); for an alternative semantics, see Nolan (1997).
6. This accommodates the enactivist point that a perturbation triggers, but does not specify, the changes it occasions in a system, which are fixed by the system's own structure (Maturana and Varela 1987).



(a) resemblance (b) pairing

FIGURE 2. Two relations that are not dependence. (a) Resemblance: two contrasts are alike. (b) Pairing: a is paired with c, and b with d. The pairing is drawn without direction, since it can be read either way.

tion as a distinction that makes a difference (MacKay 1969). The first difference is K, the second is K', and the making is the dependence. It should not be replaced by an appeal to "systematic" or "relevant" coupling, which brings in a standard of relevance or a regularity across cases.

What the dependence consists in is a matter for each domain: in physical domains it may be causal, in others logical or constitutive. The core requires that there be such a dependence, not that it be of one kind. It is admitted, not derived from the contrasts it relates: a relation cannot be obtained by describing what it relates.

Two matters are left open: whether two contrasts can each depend on the other, and whether dependence is transitive. They cannot both hold, since if dependence were transitive, two contrasts that depended on each other would each depend on itself.

A contrast may depend on several contrasts. It then stands in the same relation to each, and the core does not single out one of them.

The direction of the dependence gives two roles. In an instance of it, the field whose contrast is depended on is the *target*, and the field whose contrast depends is the *receiver*. The roles are places in the relation, not natures of what occupies them: the same field may be target in one instance and receiver in another, and one field may be both at once, through two contrasts.

5. The Receiving Side

Let R designate what bears the dependent contrast.[7] R is a field in the same sense as F: whatever bears a contrast is, insofar as it does, a field, and bearing a contrast is not something peculiar to receivers. R is not a person, a conscious

7. The vocabulary of reception recalls Kant's characterization of sensibility as receptivity, the capacity to acquire representations through being affected by objects (Kant 1998, A19/B33, cf. A50/B74). The receiver introduced here is thinner: it involves no representations, and nothing is claimed about forms of intuition.

subject, or an independently bounded system, and it may be an aspect of a larger organization that also includes F, since a distinction of roles does not establish two substances.

The contrast borne by R is between two of its *conditions*, r and r'. By the decision of Section 2 both are actual; they are not alternatives between which R is poised. It is enough that one of them is as it is in virtue of the contrast of F; the other need only be that from which it differs. Nothing requires that the receiving contrast be like the contrast of F, and no condition is assigned to a particular term. A receiver with a single condition is outside the account.

R distinguishes nothing. That its contrast holds in virtue of a contrast of F is a fact about the two contrasts, not something that R notes. It is the account that picks the contrasts out, for R as for F.

Neither contrast comes with the other. A contrast of F may hold while nothing depends on it, and R may bear contrasts that depend on nothing in F.

This fixes how much individuation the account admits. There must be enough to distinguish the two contrasts, and to take r and r' as conditions of one field. There need not be enough to identify enduring objects or successive states of a continuing subject. To assume that would build into the foundation a persisting, bounded bearer—which is what an account of a knowing subject must eventually explain.

6. Registration

definition of registration

A contrast K is registered by a contrast K' when K' holds at least partly in virtue of K. The field that bears K' registers K.

Registration is always of a particular contrast: to say that R registers F without qualification is to leave the claim incomplete. It is an actual relation between contrasts that hold, not a disposition of R to respond (Figure 3).

By (D) and (C), for something of F to appear is for a contrast of F to be registered: there is the contrast that appears, another contrast, and the dependence of the second on the first. The registering contrast is a contrast like any other, and it can itself appear, by being registered in turn. Registration is supplied by neither side alone. Keeping the three contributions distinct prevents it from being attributed to the target merely because the target is differentiated, or to the receiver merely because its conditions differ.

The definition attributes to the receiver no understanding, identification, or judgement. Where registration is realized physically, it is found in systems to which nobody would attribute knowledge, and by itself it does not place

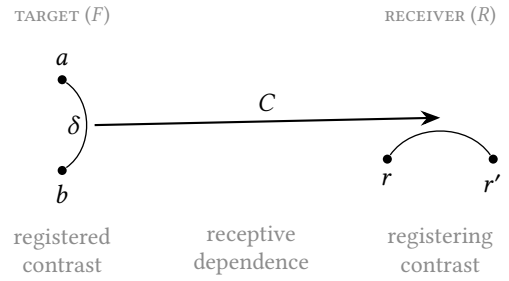


FIGURE 3. Registration. The contrast between the conditions r and r' of R holds at least partly in virtue of the contrast of a and b in δ. All four terms are actual. The registering contrast is drawn with a different organization, since it need not be like the registered one, and no term is paired with a condition. The arrow denotes receptive dependence only: it represents neither a transfer nor an order of occurrences, and the columns are roles, not regions.

anything "in the logical space of reasons".[8] It requires no transfer from F to R, no resemblance between the two contrasts, and no representational space or repertoire in R.

What is registered is always a difference between actual terms: that a differs from b. Which way a term is as against an alternative, and that two terms are alike, are not registered. Accounts of information and knowledge that begin from alternatives presuppose a set among which the actual case is singled out;[9] the core has no such set.

7. Two Cases

Two cases show the structure with dependences of different kinds.

In the first, a and b are two bodies that differ in mass, resting on the pans of a balance whose ends stand at different heights. K is the contrast of the bodies in mass, and K' the contrast of the ends in height. K' holds in virtue of K, causally and in part: it holds also because of how the balance is built and

8. The phrase is from Sellars (1956, § 36); Brandom (1994, ch. 4) develops the contrast between such reliable differential responsiveness and conceptual capacities. In perception, Dretske's non-epistemic seeing, in which an object is visually differentiated from its surroundings without any belief about it, contains a registration in the present sense (Dretske 1969).

9. Shannon's theory measures information as selection from a set of possible messages (Shannon 1948), and Dretske's informational content is defined by conditional probabilities over such alternatives (Dretske 1981). In epistemology, relevant-alternatives theories and contrastivism analyse knowledge as the elimination of alternatives (Dretske 1970; Lewis 1996; Schaffer 2004); counterfactual and interventionist analyses of dependence share the same structure (Lewis 1973; Woodward 2003).

where the bodies rest on it. The balance registers the difference in mass. It has no repertoire of the masses the bodies might have had.

In the second, δ is a fine respect and Δ a coarser one, such that terms which differ in δ in a certain way differ, for that reason, in Δ; a finer and a coarser classification of the same terms are an example. Let a and b differ in δ in that way. K is their contrast in δ, and K' their contrast in Δ. They are two contrasts with the same terms. K' holds in virtue of K, wholly, and the dependence is not causal, temporal, or spatial. One field bears both contrasts and registers one through the other.

In both cases a difference appears. In neither would anyone speak of perception or knowledge: those involve a subject, which the core does not contain.

8. Registration and Order

The core assumes no order. The dependence has a direction, but the direction of a relation is not an ordering of occurrences: that K' holds in virtue of K does not place either before the other.

The core therefore cannot say that a receiver *acquires* a contrast, *retains* it, or *compares* it with another.[10] Comparison requires that two registering contrasts both hold; it does not by itself require order. Retention requires that one receiver be identified across distinct places of some ordering. Acquisition requires that a place at which a contrast is not registered stand in an asymmetric relation to one at which it is. Neither follows from (C).

What they require is order in the loosest sense: an asymmetric relation among places, with no requirement that any two places be comparable, and with no metric, no linearity, and no universal index. Temporal succession is one way of realizing such an order; the order of steps in an inference or a proof is another, which is not temporal. The core is compatible with either and requires neither.

If receptive dependence is written with a directed symbol, such as an arrow from F to R, that symbol should not also serve, without further explanation, to represent order. The direction of dependence and the asymmetry of an ordering play different roles and should be kept notationally distinct.

10. The three operations closely parallel the synthesis of apprehension, reproduction, and recognition in the first edition of Kant's Transcendental Deduction (Kant 1998, A98–A110), and Husserl's distinction between retention and recollection (Husserl 1991) analyses the second of them. Both treat the operations as presupposing temporal form; the present account speaks only of order, of which temporal order is one kind.

9. Conclusion

Table 1 summarizes the elements of the account.

Table 1. The elements of the account, with what each permits and what each leaves unestablished.

ELEMENT	WHAT IT PERMITS	WHAT IT DOES NOT ESTABLISH
(D) Differentiation	A difference attributable to what is under consideration, by which something of it can be identified.	Which contrasts F bears; that they appear anywhere.
Actual terms only	A core without possibilities or a repertoire.	Which way a term is as against an alternative; anything of a lone term; that two terms are alike.
Specified articulation	Terms and the respect in which they differ.	Its own realization, an exhaustive partition, or objecthood.
The receiving field R	Two distinguishable receiving conditions.	A knowing subject, or any dependence on F.
(C) Receptive dependence	Registration of a contrast; the roles of target and receiver.	Understanding, representation, order, acquisition, or retention.

The proposal is that for anything to appear, what is under consideration must bear a difference, and another difference must hold in virtue of it: both between actual terms, with target and receiver distinguished as roles within that dependence.

Five questions remain open. What the dependence consists in within a given domain, and whether it can be explained by more elementary relations; to write it as an arrow would not answer this. What must be added to a registration for something to appear to a subject, that is, to be perceived. How which way a term is, and that two terms are alike, can appear without alternatives. What order acquisition and retention require. And whether what is said of things that cannot act on whoever speaks of them, things picked out by description, future events, and mathematical objects, rests on registration at all.

The account ends at registration. Order, memory, recognition, and representation have not been presupposed in reaching it, and none of them has yet been derived from it.

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